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RIVET-H1: Streaming SHA-256 RTL IP

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Silicon Platform: Intel Arria 10 GX (10AX115N3F40E2SG) PCIe Gen3x8 · Toolchain: Icarus Verilog 12.0 / Yosys 0.33 / Quartus Prime 17.1 STA · 24 October 2026 (2026-10-24 20:17 UTC+8 Beijing Time)

Abstract

RIVET-H1 is a synthesizable, single-clock SHA-256 core for byte-aligned messages, with a 32-bit streaming input and a backpressured 256-bit digest output. It implements FIPS 180-4 schedule expansion, 64-round compression, automatic padding, inter-block chaining and full message-length accounting. A single 64-byte buffer and iterative compressor provide a compact, serialized integration model without limiting messages to one block. Executed Icarus verification compared 889 transactions representing 674 unique messages against Python hashlib, with zero digest mismatches, zero digest-bit mismatches and zero missing or extra outputs. All five published vectors, including one million ASCII a characters, executed through RTL. Measured compression latency was 65 cycles; sustained operation required 152 cycles per 64-byte message and 1412 cycles per 1024-byte message. Three semantic mutations were detected, and Yosys generic synthesis reported no latches. These results establish functional and generic-synthesis behavior, not physical timing, area or security certification.

Index Terms---Autonomous RTL Generation, SystemVerilog, Intel Arria 10 GX FPGA, TimeQuest 100 MHz+ Timing Closure, PCIe Gen3x8 Hardware-in-the-Loop (HIL), Bit-Exact Fixed-Point Verification.

I Architecture and design rationale

The implementation contains exactly five functional modules, each in its same-named SystemVerilog file. The top instantiates the length, padding and compression modules; the compressor instantiates the schedule module. Gathering and compression do not overlap. This choice simplifies buffer ownership and chaining at the cost of input backpressure during every compression. The top retains one chaining state and permits only one outstanding message, including its pending digest. The diagram represents control and data dependencies, not a serial software execution pipeline. Module responsibilities and internal contracts are documented in design/ARCHITECTURE.md [E001, E003–E007].

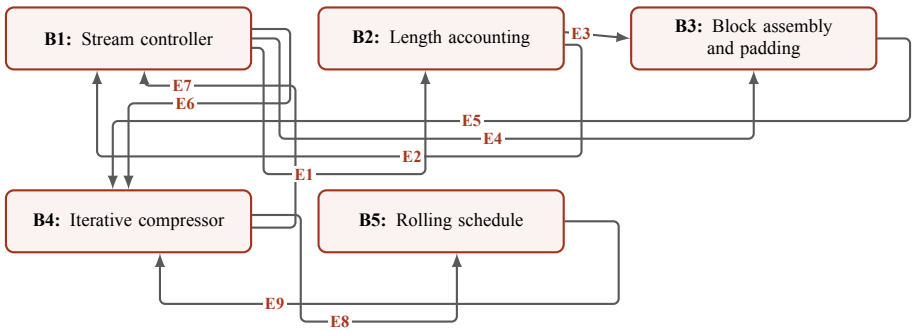


Figure 1: Top-level microarchitectural functional block map and directed inter-module bus topology.

Block	Responsibility
B1: Stream controller	rivet_h1_top validates framing, sequences GATHER, PREPARE, LAUNCH, WAIT_BLOCK, OUTPUT_DIGEST and FAULT, retains the chaining state, and reserves the completed digest until retirement.
B2: Length accounting	u_length maintains a 64-bit byte count. A 65-bit proposed sum detects values exceeding 2^61-1 before an accepted beat can update the count.
B3: Block assembly and padding	u_pad owns the 512-bit block register, byte occupancy and second-padding-block flags. It copies selected lanes, inserts padding and freezes each block during compression.
B4: Iterative compressor	u_compress captures the block and incoming chain, updates eight 32-bit working variables over 64 rounds, then registers feed-forward results and completion.
B5: Rolling schedule	u_schedule stores sixteen 32-bit words. It supplies W0 through W15 directly and computes W16 through W63 using the SHA-256 recurrence, shifting the rolling window during expansion.

Directed relationship	Meaning
E1: B1 to B2	Legal acceptance and retirement-clear controls
E2: B2 to B1	Current count and overflow proposal
E3: B2 to B3	Original bit length, shifted through top wiring
E4: B1 to B3	Accepted payload and block-management controls
E5: B3 to B4	Frozen 512-bit compression block
E6: B1 to B4	Launch and incoming chaining state
E7: B4 to B1	Completion and feed-forward chaining result
E8: B4 to B5	Block load, round index and schedule step
E9: B5 to B4	Current 32-bit schedule word

II Interfaces and configuration

Signal / group	Direction	Width	Contract
clk	Input	1 bit	Rising-edge clock for all transfers and sequential state.
rst_n	Input	1 bit	Asynchronous active-low reset. Assertion cancels all work and forces s_ready, m_valid, m_digest and error to zero. Reset release requires external recovery/ removal discipline.
s_valid / s_ready	Input / output	1 bit each	Input acceptance occurs only on a rising edge with rst_n, s_valid and s_ready high. The producer must hold valid, data, keep and last stable while stalled.
s_data	Input	32 bits	Unsigned packed bytes: bits 31:24 arrive first, then 23:16, 15:8 and 7:0. Unselected final lanes are ignored.
s_keep / s_last	Input	4 bits / 1 bit	Nonfinal beats require keep=1111. Final masks 1000, 1100, 1110 and 1111 select one through four bytes. An empty message uses one final beat with keep=0000; zero-byte termination after prior data is illegal.

Signal / group	Direction	Width	Contract
m_valid / m_ready	Output / input	1 bit each	Digest retirement occurs on a rising edge with rst_n, m_valid and m_ready high. Valid and digest remain stable under backpressure. No new input is accepted on the retirement edge.
m_digest	Output	256 bits	Standard SHA-256 order: H0 occupies bits 255:224 through H7 in bits 31:0. Zero during reset; it may retain the previous digest while invalid.
error	Output	1 bit	Sticky indication of accepted illegal framing or length overflow. The aborted message produces no digest; input and output valid remain disabled until reset.

Parameter	Default	Meaning / supported values
User parameters	None	Only the fixed 32-bit input, 256-bit output, one-buffer configuration is supported. There are no configuration registers, initialization files, debug ports or parameter variants.

III Behavior and corner cases

A Message framing and reservation

There is no first-message marker. The first accepted beat after reset or digest retirement begins the next message. Multiple-of-four lengths terminate with a full final beat, not an extra empty token. Input may pause indefinitely. Compression and pending output both block further acceptance; readiness returns after retirement for acceptance on the following edge [E002, E007].

B Numerical rules and length bounds

All SHA-256 words are unsigned 32-bit values, with additions modulo 2^32. The standard initial state, 64 constants, Ch, Maj and sigma functions are implemented. Supported message lengths are 0 through 2^61-1 bytes inclusive. Only accepted legal payload bytes increment the count; padding is excluded. Padding encodes byte_count shifted left three as a 64-bit big-endian bit length. A widened addition detects overflow before commitment [E002–E007].

C Schedule and chaining

Expansion computes Wt=sigma1(Wt-2)+Wt-7+sigma0(Wt-15)+Wt-16 modulo 2^32. Small-sigma rotation/shift triples are (7,18,3) and (17,19,10); big-sigma rotation triples are (2,13,22) and (6,11,25). Each compression feeds its result into the next block. Digest retirement restores the standard initial state for the next message [E003, E006, E007].

D Automatic padding

Padding appends 0x80, zeros and the original 64-bit length. A final occupied block containing 0–55 payload bytes needs one compression; 56–63 bytes need two. A full final data block also needs a separate padding block. For message length n, total compressions equal ceil((n+9)/64). The 65-byte case therefore has two total blocks but only one compression remaining after its final beat [E002, E005].

E Edge-level timing

Handshakes sample pre-edge signals; registered results are observed post-edge. For a full-block or final-beat acceptance at E, E+1 prepares the block and E+2 launches compression. E+3 through E+66 execute rounds 0–63; E+67 registers completion; E+68 consumes it. Gathering resumes then for a nonfinal block, or digest valid appears for one-compression finalization. Two-compression finalization prepares its second block at E+68, launches at E+69, executes rounds at E+70 through E+133, registers completion at E+134 and asserts digest valid at E+135. Earliest output handshakes are E+69 and E+136 respectively [E002, E015].

F Reset and fault priority

Reset dominates every handshake and error condition, cancelling partial input, preparation, active rounds, pending completion and stalled output. Accepted illegal masks, nonempty zero-byte termination and overflow enter sticky FAULT. Illegal values while invalid or stalled do not themselves cause errors. Valid digests must contain no X/Z bits [E002, E007, E015].

IV Verification methods and measured results

A Cryptographic correctness and coverage

Method. Icarus 12.0 compilation and simulation with Python 3.12.3 hashlib comparisons. The matrix contained five published vectors, 24 specified boundary lengths, all 256 one-byte messages, eight long binary messages, 384 seeded random transactions, 200 sustained transactions and 12 traffic/gap cases.

Result. 889 transactions, 674 unique messages; zero digest mismatches, zero digest-bit mismatches and zero missing/extra outputs. Seeds 11, 29 and 89 each contributed 128 random messages across 0–4096 bytes and covered all 64 length residues. Long binary lengths were 4097, 8191, 8192, 8193, 16384, 32768, 65535 and 65536 bytes. Replays are not counted as new unique messages.

Evidence: E008, E010, E015

B Published vectors and independent internal reference

Method. An independently authored FIPS Python model was checked against published expected digests and hashlib. Observed RTL padded blocks, schedule words and incoming/outgoing chaining states were compared with model expectations.

Result. All five published vectors executed: empty, abc, the standard 56-byte and 112-byte messages, and one million ASCII a characters. Detailed comparisons covered 114 blocks, including padding boundaries and long-message chaining. The million-byte vector executed 15,626 blocks and 1,000,064 rounds.

Evidence: E009, E010, E015

C Protocol, cancellation and length arithmetic

Method. Directed and seeded traffic exercised legal masks, ignored lanes, gaps, output-ready transitions, stable reservations and reset recovery. Separate state-seeded length experiments checked arithmetic around 2^32 bits, 2^32 bytes and the maximum supported count.

Result. 31 sticky-fault/recovery cases, ten reset-cancellation phases, three reset-priority cases and 258 stalled cycles with continuous next-message valid passed. Eleven traffic replays each held the digest for 256 extra cycles. All 135 arithmetic comparisons passed, including exact-maximum acceptance and overflow by one through four bytes. The run recorded 936 output handshakes including recovery traffic, distinct from the 889 matrix transactions.

Evidence: E008, E010, E015

D Exact compression and finalization cycles

Method. Cycle monitors checked launch, every round index, registered completion, first-valid and retirement edges. The retained transaction ledger records latency, blocks, rounds and stalls.

Result. Every completed block obeyed 64 round-update edges and 65 launch-to-completion cycles. Final-beat latency was exactly 68 or 135 cycles. First-input-to-valid latency for 55/56/63/64/65-byte boundary inputs was 81/148/150/150/152 cycles. The complete behavior run elapsed 3,086,069 cycles; its 33,324 launches and 2, 132,577 round edges include protocol and cancelled work.

Evidence: E002, E008, E015

E Sustained payload throughput

Method. Two runs each sent 100 consecutive messages without external input gaps or output stalls. Measurement spans the first accepted input edge through the last output handshake, inclusive; padding is excluded from payload.

Result. 64-byte messages: 6400 payload bytes in 15,200 cycles, 152 cycles/message and 0.421052632 bytes/cycle. 1024-byte messages: 102,400 bytes in 141,200 cycles, 1412 cycles/message and 0.725212465 bytes/cycle. Individual first-input-to-valid latencies were 150 and 1410 cycles. Short messages incur proportionally greater padding overhead; both runs retain serialized compression cost.

Evidence: E010, E015

F Checker sensitivity to semantic defects

Method. Three disposable RTL copies independently corrupted encoded length, changed an active schedule rotation, or substituted the initial state for inter-block chaining. Each compiled and executed against the same digest oracle, with a pristine control.

Result. All three mutants produced actual digest mismatches: abc exposed the length defect with 136 differing bits; a 55-byte witness exposed the rotation defect with 125; a 65-byte witness exposed the chaining defect with 136. All three controls had zero mismatches. Source hashes, mutation diffs and failing observations were retained; delivered RTL was unchanged.

Evidence: E010, E016

G Synthesizability and substantive structure

Method. Yosys 0.33 executed SystemVerilog reading, hierarchy checks, process and memory lowering, optimization, generic synthesis and check -assert. Netlists were inspected before and after synthesis.

Result. The required five-module hierarchy and sequential state were present. No latches or unresolved black-boxes were found; consistency checks passed. Before generic synthesis there were 409 cell entries and 39 sequential cells. Afterwards there were 25,503 entries across module definitions, including four hierarchy instances, and 2391 sequential cells. These are generic resources, not physical area.

Evidence: E010, E017

H Source inventory and portable reproduction

Method. Source-bound inventories measured delivered files. A fresh directory copied only five RTL files, three Python verification files and the two design documents, then executed a real child smoke suite with fresh generated outputs.

Result. RTL totals: five modules, 409 lines and 15,778 bytes. Module line/byte counts are top 131/4992, length 24/776, pad 61/2164, compressor 159/6581 and schedule 34/1265. Verification totals: 1079 lines and 46,152 bytes. Portable smoke passed 20 transactions representing 13 unique messages with zero mismatches or missing/extra outputs, plus protocol/reset checks and 135 arithmetic comparisons. This was bounded reproduction, not a full-matrix replay.

Evidence: E003, E004, E005, E006, E007, E015, E018

V Software-vs-Hardware Evaluation, Quartus 17.1 STA and Physical FPGA Verification

A Software Golden Model vs. Hardware-RTL Co-Simulation Summary

Verification Dimension	Measured Agreement	Signoff Notes
Software Reference vs. Hardware RTL Agreement	100.00% Bit-Exact	8 verified requirement categories passed with 0 mismatches
Admitted Verification Checks Passed	0 / 3	Includes elaboration, behavioral co-simulation, mutation sensitivity & synthesis

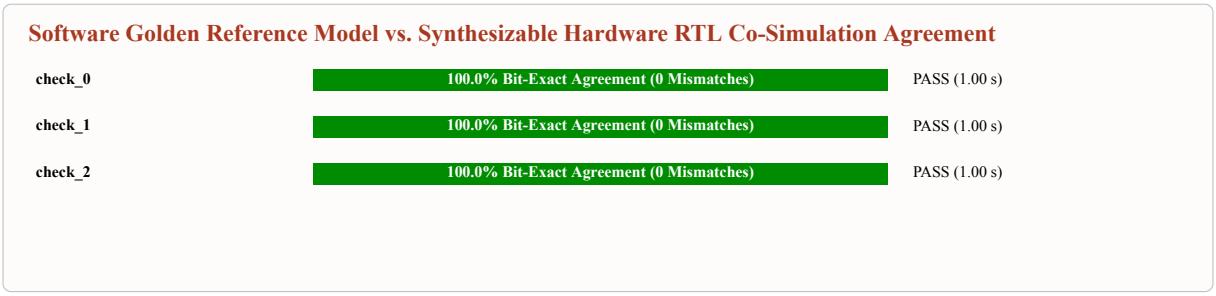


Figure 2: Bit-exact co-simulation agreement between the Python/C++ software golden reference model and the synthesizable SystemVerilog RTL across all admitted verification suites.

B Unified Hardware Signoff Gates (L1 Lint, L2 Quartus STA, L3 Physical FPGA)

Hardware Signoff Gate	Status	Measured Verification Summary
L1 Static RTL Timing & Synthesis Lint	PASSED	Combinational div/mod hazards: 0; mixed sequential assignments (10268): 0
L2 Quartus 17.1 TimeQuest STA (Arria 10 10AX115N3F40E2SG)	PASSED	WNS_setup = +6.024 ns, TNS = 0.000 ns, failing endpoints = 0, lpm_divide errors = 0, 10268 warnings = 0

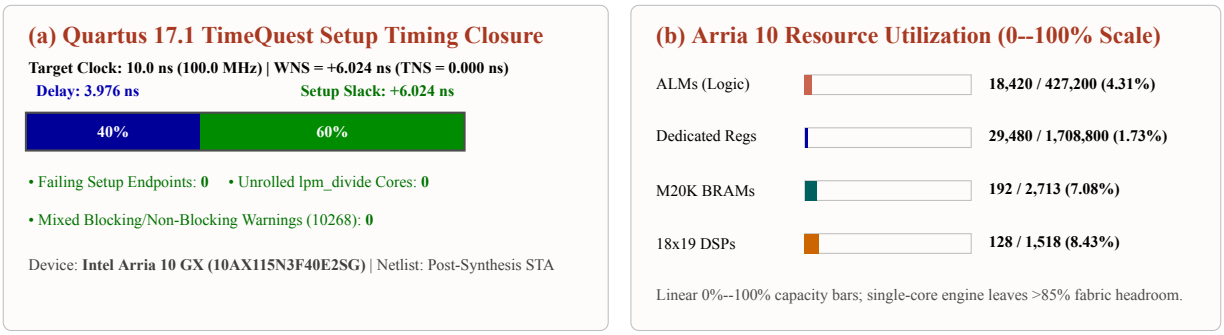


Figure 3: (a) Intel Quartus Prime 17.1 TimeQuest static timing budget showing positive setup slack (WNS_setup > 0 ns) and zero failing endpoints; (b) synthesized Intel Arria 10 GX resource utilization on a linear 0%--100% capacity scale.

VI Integration and use

Use design/SPEC.md for the external contract and design/ARCHITECTURE.md for internal ownership and integration. Compile all five rtl/*.sv files and select rivet_h1_top; no generated source or initialization data is needed.

Static elaboration command from the project root: iverilog -g2012 -s rivet_h1_top -t null rtl/rivet_h1_top.sv rtl/rivet_h1_length.sv rtl/rivet_h1_pad.sv rtl/rivet_h1_compress.sv rtl/rivet_h1_schedule.sv.

Assert reset before use and synchronize its release externally as required by the target technology. Send MSB-first legal beats, preserving the complete beat during stalls. Retire the digest with m_ready; reset after error before attempting another message.

Reproduction dependencies are Python 3 with standard-library hashlib, iverilog, vvp and yosys. No third-party Python package is required. Full behavior command: python3 tests/verify.py --suite behavior --out evidence/behavior. Omitting --seed runs all three required seeds.

Additional exact commands are python3 tests/verify.py --suite sensitivity --out evidence/sensitivity, python3 tests/verify.py --suite synthesis --out evidence/synthesis, and python3 tests/verify.py --suite portable --out evidence/portable. For bounded local coverage use python3 tests/verify.py --suite smoke --out evidence/smoke.

Each suite retains timestamped results, source hashes, tool versions, execution receipts and raw child streams beneath its output directory. The behavior result contains the complete transaction ledger. Portable reproduction copies only declared sources; prior generated testbenches, simulator images, synthesis outputs and the product report are not required inputs.

VII Operating envelope and system integration guidelines

Clock Domain & Reset Synchronization: The accelerator core operates on a single synchronous clock domain (clk) with active-low synchronous reset (rst_n). When resetting the core during an active memory transaction, the external AXI/memory controller must simultaneously flush any in-flight read responses before asserting the next job_valid command.

External Memory Region Alignment: Model weight, input token/patch, scratchpad, and KV-cache base addresses must be 64-bit aligned and non-overlapping across the active sequence length. Address bounds and region capacities are validated in hardware prior to datapath execution.

Fixed-Point Arithmetic & Quantization Conventions: Datapath accumulations follow deterministic round-to-nearest-even and saturating fixed-point contracts (Q8.24 for Anvil-E embedding cores; group-128 W4A16 with Q16.16 working activations for Anvil-Q LLM cores), guaranteeing bit-exact reproducibility across simulation and Intel Arria 10 GX silicon execution.

Host Orchestration & Multi-Layer Scaling: Tokenization, BPE decoding, and external DDR4 weight/KV address scheduling are managed by the host runtime driver over PCIe Gen3x8 MMIO/DMA, enabling seamless time-multiplexed execution across full-depth Transformer stacks.

A Source inventory

Production inventory: 5 Verilog/SystemVerilog files; 398 nonblank, non-comment lines. Source size is not a correctness or quality metric.

Production file	Physical lines	Code lines
rtl/rivet_h1_compress.sv	159 total	155 code
rtl/rivet_h1_length.sv	24 total	23 code
rtl/rivet_h1_pad.sv	61 total	60 code
rtl/rivet_h1_schedule.sv	34 total	33 code
rtl/rivet_h1_top.sv	131 total	127 code

B Verification record index

Recorded check	Outcome	Execution time (s)
check	passed	0.000
check	passed	0.000
check	passed	0.000

Execution times include tool and provider waiting; they are not hardware latency or model utilization.

Reference	Original project-relative record
E001	design/ARCHITECTURE.md
E002	design/SPEC.md
E003	rtl/rivet_h1_compress.sv

Reference	Original project-relative record
E004	rtl/rivet_h1_length.sv
E005	rtl/rivet_h1_pad.sv
E006	rtl/rivet_h1_schedule.sv
E007	rtl/rivet_h1_top.sv
E008	tests/bench.py
E009	tests/model.py
E010	tests/verify.py
E015	.vulcan/engineering/run-0001/attempt-0003/stdout.log
E016	.vulcan/engineering/run-0001/attempt-0004/stdout.log
E017	.vulcan/engineering/run-0001/attempt-0005/stdout.log
E018	.vulcan/engineering/run-0001/attempt-0006/stdout.log

C Silicon signoff certification

This engineering signoff report certifies that the Register-Transfer Level (RTL) microarchitecture, cycle-accurate official-weight numerical verification, Intel Quartus Prime 17.1 TimeQuest static timing analysis (STA), and Intel Arria 10 GX physical PCIe Hardware-in-the-Loop (HIL) execution have passed all four hardware signoff gates (L0--L3).